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PERSONAL SUMMARY

Machine Learning Engineer who has always tried to be **statistically** perfect, focusing not just on building models, but on understanding and how to optimize them so they deliver genuinely useful insights to the end stakeholder. You'll find me either breaking models purposely to understand or reading research papers. Currently serving my dissertation(In **medical Imaging**, trying to **restore** the images using **GenAI models** and comparing the **explainability** shift across models) period at Dublin City University where in, I did MSc in Computing with a specialization in Artificial Intelligence with internship experience, hackathon wins, and a hands-on foundation in data and ML. Someone who understands the problem before jumping to a solution.

EDUCATION

MSc in Computing (Artificial Intelligence) (Expected 1:1) – Dublin City University, Ireland [2025 – 2026]

- Modules include **Neural Language Modelling, Statistical Modelling, Data Visualisation, LLM and Reinforcement Learning.**

B.E. in Artificial Intelligence and Data Science (CGPA: 7.58/10) – TEC, Navi Mumbai [2021 – 2025]

- Core coursework: **Database Management, Statistics & Probability, Data Mining, Supervised and Unsupervised ML, Reinforcement Learning, Deep Learning.**

KEY SKILLS

Data Analysis & BI: Excel, Power BI, data validation, cross-source data reconciliation, dashboard development, data storytelling.

Programming & Databases: Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn), SQL, R.

ML & DL: Supervised Learning, NLP, Pytorch, scikit-learn, XGBoost, LightGBM, CNNs, Vision Transformers, Diffusion Models.

GenAI & LLM's: RAG, Embeddings, vector databases, Prompt Engineering, LLM Agents, Multimodal pipelines.

Cloud & Deployment: AWS (S3, SageMaker, Lambda, SNS), FastAPI, Django, Docker.

Soft Skills: Clear communication of analytical findings to non-technical stakeholders, team collaboration, attention to detail, problem-solving.

RESEARCH EXPERIENCE

Towards a Robust and Explainable Pipeline for Diabetic Retinopathy Classification through Quality-Aware GenAI Image Restoration | MSc Dissertation under Supervisor: Prof. Alessandra Mileo | [Github](#)

🏆 **Accepted Poster — OMIA Workshop, MICCAI 2026, France**

- Designed a controlled study measuring how diabetic-retinopathy diagnostic accuracy and model explanations behave under nine synthetic degradations (blur, exposure, and noise at three severity levels) on the APTOS 2019.
- Benchmarked CNN's against a Vision Transformer (ViT-Base); found ViTs substantially more robust under severe noise (~62% vs ~20% accuracy) a gap invisible on clean benchmarks and showed Explanations(Grad-CAM, SHAP, Attention Rollouts) drift faster than accuracy as quality drops demonstrating that a model can be right for the wrong reasons.
- Evaluated GenAI restorers(CLAHE, A-ESRGAN, SwinIR+GAN, Cold Diffusion, Conditional DDPM) and established that pixel fidelity gains (e.g. Cold Diffusion, +12 dB PSNR) do not improve and often harm downstream diagnosis, due to classifier distribution shift.

PROFESSIONAL EXPERIENCE

ML/AI Engineer | VenueFlow.ie | Dublin, Ireland | May 26 – Present

- Currently building an autonomous AI wedding-coordination system for the Build with Gemini XPRIZE (Aug 2026); designed four Gemini-powered agents (intake, planning, venue/supplier matching, quote extraction) with structured outputs, hallucination guardrails, budget validation, and audit logging, backed by 29+ automated tests for guardrails, chain tracing, and error rollback.

Data Science Engineer | Bharat Intern | India | Dec 2023 – Feb 2024

- Engineered an SMS spam classifier (TF-IDF + SVC) reaching 91% precision-recall on 5,500+ messages (~40% fewer manual reviews), and a CNN image-classification pipeline (TensorFlow/Keras) with augmentation, dropout, and early stopping that improved generalisation by 12% on a ~2,000-image dataset.

Python Developer Intern | Mira Advance Engineering | India | Sep 2023 – Nov 2023

- Led employee/order management system (SQLite, admin and employee dashboards) with robust data validation and exception handling, improving operational speed 2x and cutting data errors by 30%.

KEY PROJECTS

AI Generated Image Detection for Marketplace Integrity | Etsy ML Challenge | [Github](#)

Invited to present at **Etsy HQ** — **Won 2nd Place**

- Developed a classifier separating authentic photographs from AI-generated (diffusion / GAN) images on a ~6,200-image marketplace dataset, reaching 0.941 validation F1 via Optuna-tuned weighted ensembles.
- Discovered that intermediate CLIP transformer layers (18–23) substantially outperform final-layer features (0.923 vs 0.848 F1), then fused complementary semantic (CLIP, DINOv2) and low-level frequency / compression signals (FFT, ELA) for the performance lift.
- Applied 5-fold cross-validation, test-time augmentation, and conservative pseudo-labeling; presented findings at Etsy's office following a direct invitation based on the research.

3D Spatial Reasoning (Nomad AI) - LLM + Geometry Engine | [Github](#)

Neuro-symbolic system for collision-free, explainable object placement

- Built a system for room-aware 3D object placement that avoids LLM "coordinate hallucination" by separating responsibilities: the LLM parses natural-language intent into structured constraints, a deterministic geometry engine computes collision-free candidates (AABB checks, clearance and walkway rules), and the LLM explains the final choice with measured distances.
- Parsed Blender scene geometry (boundaries, obstacles, doorways), generated and ranked candidates by constraint satisfaction, and returned physically valid, human-readable placements.

HealEdge - Federated Learning for Diabetic Retinopathy Detection using XAI | [Github](#) | 3rd place Advanced HealthTech and MedTech Innovation Hackathon, Trinity College Dublin

- Addressed a federated learning framework training a ResNet-50 pipeline across distributed, non-IID hospital datasets without centralising sensitive retinal scans (GDPR/HIPAA-compliant), managing communication overhead to maintain convergence.
- Integrated **Grad-CAM** explainability to generate visual heatmaps of model predictions, enabling clinicians to verify which retinal regions influenced the diagnosis a critical requirement for trust and adoption in clinical decision-support systems.

VoxSight : Real-time Multimodal Vision Copilot | In development | [Github](#)

- Building a real-time accessibility copilot that streams camera + voice over WebSocket to a vision-language model and returns spoken scene descriptions (STT → vision LLM → TTS).
- Designed a swappable provider architecture (mock/live per service) and a deterministic routing layer for turn-by-turn navigation, keeping spatial guidance model-free to cut latency and cost. Separated LLM intent-parsing from deterministic execution — the same neuro-symbolic pattern used to prevent coordinate hallucination in prior work.
- Conducting active literature research for **VoxSight Recall**: studying episodic visual memory (MemGPT, Ego4D, Neural Turing Machines, RAG) and designing an event-schema-based memory layer over EPIC-KITCHENS-100 to distinguish *last-seen* from *current* object location — addressing a safety-critical gap where existing systems give confidently wrong answers to blind/low-vision users. Reproducing Lewis et al. RAG and SteerViT as baselines. [[Research Notes](#)]

ACHIEVEMENTS

- **Accepted Poster Presentation** at the **OMIA Workshop, MICCAI 2026, France** — *"Towards a Robust and Explainable Pipeline for Diabetic Retinopathy Classification through Quality-Aware GenAI Image Restoration"*
- **Won 3rd place out of 70 teams** – Advanced HealthTech and MedTech Innovation Hackathon, Trinity College Dublin – Federated Learning for Diabetic Retinopathy detection. [2026]
- **Won the Motivalogic Hackathon** sponsored by AWS (Dublin) in collaboration with Cloudhight Consultancy, leading a cross-functional team to deliver a CPU log anomaly detection solution for fraud activity. [2025]
- Pitched research paper (Autonomous Drone Navigation) at the **International Conference on Engineering Research and Innovations**. [2025]

CERTIFICATIONS & ACTIVITIES

- ML, Python, Flask, RAG.
- **Ambassador, Thinking About Thinking** – responsible-AI community building (AE Global Summit, London 2026)